7 December 1982
SUPERSEDING
MIL-A-28988A(MC)
14 February 1973

MILITARY SPECIFICATION

AMPLIFIER, RADIO FREQUENCY: AM-6545/GRC-193

This specification is approved for use by the Marine Corps, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for the design, fabrication and testing of one type radio frequency (rf) amplifier.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704	-	Treatment and Painting of Materiel.
MIL-C-5809	-	Circuit Breaker, Trip-free, Aircraft, General Specification for
MIL-C-12520	-	Connector, Plug and Receptacle (Electrical, Waterproof), and Accessories, General Specification for
MIL-M-13231	-	Marking of Electronic Items.
MIL-P-15024	-	Plate, Tags and Bands for Identification of Equipment

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-E-16400	-	Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-E-52798	-	Enamel, Alkyd, Camouflage
MIL-C-55181	-	Connector, Plug and Receptacle, Intermediate (Electrical) (Waterproof) General Specification for

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-242	-	Electronic Equipment Parts (Selected Standards).
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electronmagnetic Emisson and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STd-463	-	Definition and System of Units, Electromagnetic Interference Technology.
MIL-STD-470	-	Maintainability Program Requirements (for Systems and Equipments).
MIL-STD-471	-	Maintainability Demonstration.
MIL-STD-781	-	Reliability Tests Exponential Distribution.
MIL-STD-785	-	Reliability Program for Systems and Equipment Development and Production.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-831	-	Test Reports, Preparation of
MIL-STD-1281	-	Internal Transient Control For Solid State Power Supplies.
MS3114	-	Connector, Receptacle, Electric, Series 1, Solder Type, Jam Nut Mounting Nut, Bayonet Coupling.

HANDBOOKS

MILITARY

MIL-HDBK-472	-	Maintainability Predication.

(Copies of specifications, standards and handbooks, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Prior to submission of any complete production items for acceptance, the contractor shall furnish ten first articles for determination of conformance to this specification. Examination and tests shall be those spcified herein. Approval of the first articles by the procuring activity shall not relieve the contractor of his obligation to supply amplifiers conforming to this specification. Any changes or deviations of the production items from the first articles shall be subject to the approval of the contracting officer.

3.2 Parts, materials and processes. Parts, materials and processes approved in MIL-E-16400 shall be used unless otherwise directed by the procuring activity. The selection of the class, grade or type shall be such that when mounted in the equipment, the part or material will perform its intended function. Parts shall be selected from MIL-STD-242 wherever possible. All nonstandard parts shall be approved by the procuring activity before delivery of the amplifiers.

3.3 Design and construction. The amplifier described herein shall contain all of the necessary circuitry, including system control circuits and an integral power supply, to function as a transmitter with the performance as specified herein. The amplifier shall be constructed in the form of a case-chassis assembly (including handles and the finned external heat exchanger), cover and plug-in modules and printed circuit boards. As part of the vehicular AN/MRC-133 Radio Set, the amplifier shall be electrically and mechanically compatible and shall operate with the following ancillary equipment:

(a) Coupler, Antenna CU-2064/GRC-193

(b) Receiver Transmitter RT-1209/URC

(c) Mounting, Electrical Equipment MT-4495/GRC-193

(d) Mounting, Electrical Equipment MT-4497/GRC-193

(e) 15-foot whip antenna and 35 foot whip antenna

(f) Antenna AS-2259

(g) Cable, Direct Current Power CX-( )/GRC-193

(h) Cable, Control CX-( )/GRC-193 (AM-6545 to RT-1209/URC interconnection)

(i) Cable, Control CX-( )/GRC-193 (AM-6545 to CU-2064 interconnection)

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3.3.1 Modes of transmission. The amplifier shall provide the emission of single sideband (ssb) (upper sideband (usb) and lower sideband) voice, tone keyed continuous wave (cw), compatible amplitude modulation (am), frequency-shift keying (fsk) and narrow band secure voice signals over the frequency range of 2.0000 to 29.9999 megahertz (MHz) in 100 hertz (Hz) steps generated in the RT-1209/URC Receiver-Exciter.

3.3.2 Power sources. The amplifier shall operate with any standard direct current (dc) vehicular power supply (+26.5 direct current volts (Vdc) nominal voltage) or any other source providing +26.5 Vdc nominal. A diode shall be incorporated which will protect the electronic circuitry and internal wiring if the polarity of the power source is reversed from the normal condition where the negative side is grounded.

3.3.3 Power output. The rf power delivered to a 50-ohm load shall be 400 watts (W) peak enevlope power (pep) or average.

3.3.3.1 Power output vs operating conditions. The power output in 3.3.3 shall be maintained within ±1 decibel (dB) over the 2.0000 to 29.9999-MHz frequency range, over the -40° C to +55° C ambient temperature range and over an input voltage range of 24 to 30 Vdc when the amplifier is operated with the RT-1209/URC Receiver-Exciter (closed automatic level control (ALC) loop).

3.3.3.2 Power output, continuous operation. The power output shall be maintained within the limits of 3.3.3.1 for continuous 24-hour key-down operation in any selected mode.

3.3.3.3 Warm-up time. The amplifier shall be capable of providing rated power output as specified in 3.3.3 after the 60 ±10-second warm-up delay and immediately following automatic tune cycle.

3.3.3.4 Tune power output. The amplifier shall develop sufficient rf power to properly activiate and tune the CU-2064/GRC-193 Antenna Coupler.

3.3.4 Drive level. The amplifier shall develop rated power output within the limits of 3.3.3.1 with no more than 100 milliwatt (mW) drive level from the exciter.

3.3.5 Input power. When operated from a 26.5-Vdc supply, the input power shall not exceed 1,190 W in any mode of operation.

3.3.5.1 Over-drive. Input levels up to 2-W pep shall cause no damage to the circuits in the amplifier.

3.3.5.2 Input impedance. The input impedance shall be 50 ohms with a voltage standing wave ration (vswr) not greater than 1.5:1.

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3.3.5.3 Input voltage. The amplifier shall meet requirements specified herein with input voltages between 24.0 and 30.0 Vdc. With input voltages between 22.0 and 24.0 Vdc, operation at reduced performance levels shall be possible without permanent damage or deterioration to components or circuits. To this end, all dc supply voltages except filament, plate and relay voltages within the amplifier shall be regulated to within ±3 percent when the loads to the dc supply lines vary over their extreme range and the input voltage to the amplifier varies from 22 to 30 Vdc. This requirement shall be met over the full ambient temperature range of the amplifier. The filament voltage shall be regulated not to exceed 26.5 Vdc under any operating conditions.

3.3.5.4 Tune cycle. During the initial tune cycle, when operated from a 26.5-Vdc supply, the input current shall not exceed 35 amperes (A).

3.3.6 Internal transient control. The design of the amplifier shall be such as to avoid internal transients which could result in degradation or catastrophic failure of the amplifier components. The amplifier shall not produce voltage, current or power transients during the course of operation, which affect the life of the semiconductor complement or create an unsafe condition. Internal transient control shall be in accordance with MIL-STD-1281.

3.3.7 Spurious outputs. The level of emissions other than the desired output frequencies shall be in conformance with the following, when the amplifier is loaded into a 50-ohm resistive load at the type N output connector terminal:

(a) Third order intermodulation distortion products shall be at least 30 dB below either of two equal amplitude tones separated in frequencey by 1 kilohertz modulating the transmitter to rated pep. At ambient temperatures below -25°C and with a supply voltage of 24 Vdc, degradation of up to 5 dB is permitted.

(b) All harmonically related signals shall be attenuated at least 35 dB.

(c) all nonharmonic spurious outputs shall be at least 60 dB down from the desired output signal level.

(d) Hum and noise output - Hum, ripple and other extraneous noise shall be at least 40 dB below the output level of either of two equal amplitude tones modulating the transmitter to rated pep output.

3.3.8 Frequency range. The amplifier shall be capable of operating and automatic tuning at any frequency in the 2.0000- to 29.9999-MHz range

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3.3.9 Tuning. Tuning at any frequency in the 2.0000- to 29.9999-MHz range shall be fully automatic, initiated either by switching dc power on or changing any frequency selector digit on the exciter, except 100-Hz steps. Final automatic tune cycle shall be initiated by momentary push to talk (PTT) switch closure. The automatic tune cycle shall be completed in less than 5 seconds after PTT switch closure.

3.3.9.1 Tune-up ready. The amplifier shall generate a tune-up ready signal to be "ANDED" with the coupler ready in order to indicate a condition to permit full power operation. Receipt of coupler ready shall not be utilized to generate an amplifier ready.

3.3.9.2 Homing of elements. Homing of the tuning element shall be initiated upon receipt of a single rechannel pulse. During this action, the rf shall be removed from the tuning element.

3.3.10 Receive only mode. The primary dc power connections and system control shall be such that receive operation is possible without energizing the amplifier or antenna coupler. Receive operation shall be possible when the amplifier circuit breaker has tripped. In this mode, the amplifier circuit shall be by-passed for the incoming rf signal with attenuation less than 1 dB (50 ohm input and output terminations).

3.3.11 Single tube operation. Single tube operation shall be possible in case of filament burn-out in one tube without causing secondary failures or permanent degradation of performance characteristics. An external warning lamp shall be provided to indicate a filament failure (see 3.3.29.1(j). In addition, internal test meter positions shall be provided to allow quick isolation of the faulty tube.

3.3.12 ALC. When operated with the RT-1209/URC (or equivalent) Receiver-Exciter, the amplifier shall be capable of providing performance specified herein with input levels of 250 mW +3 dB without adjustments. Operation with input levels up to 2 W shall be possible after simple ALC adjustments in the amplifier.

3.3.13 External interfaces. The amplifier shall interface with the RT-1209/URC Receiver-Exciter and the CU-2064/GRC-193 Antenna Coupler in accordance with Tables I and II and the following.

3.3.13.1 Amplifier to receiver-exciter interface.

(a) The amplifier shall accept and operate with the following control input signals from the receiver-exciter:

(1) Two binary coded 10-MHz code signals ("0" ground through a diode, "1" open circuit).

(2) Four binary coded 1-MHz code signals ("0" ground through diode, "1" open circuit).

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TABLE I. - AMPLIFIER TO RECEIVER INTERFACE

RT-1209/URC Receiver/Exciter				AM-6545/GRC Amplifier	
Line Conditions/Status	UN3026/D00 Pin Connections	Function	Signal Source Pin Connection	Line Condition/Status	Operating Limits
0-Logic-Hard and Thru Diode and Switch Contacts /200 ohms Maximum 1-Logic-Open	L M N P R S	Frequency Code, 1 MHz (0) Frequency Code, 1 MHz (4) Frequency Code, 1 MHz (2) Frequency Code, 1 MHz (1) Frequency Code, 10 MHz (2) Frequency Code, 10 MHz (1)	A B C D E F	Max "0" Current - 1MA 4 VDC pull voltage thru 15000 ohms Under Logic "1" Condition	Frequency Decoder Circuits Shall Operate Properly with the Following Limit Inputs: 0-GND thru 500 ohms Max 1-50,000 ohms Min to GND
Sidetone-2ma * x sink current Sidetone +0 VDC thru 10Kohm Pullup	P	Sidetone	H	GND thru Collector. Emitter Junction of Saturated Negative Positive Negative Transistor and 560 ohms When TX-RF Output Present. Open Collector-Emitter Junction in RX and TX (No signal)	
TIP-2ma max sink current TIP +8 VDC thru 10 K Pullup	R	Tune in Progress (TIP)	J	GND Thru C-E Junction of Resistor and 560 ohms during Auto Tune Cycle Open C-E Junct. in Normal RX-TX Operation	
Keyed thru Diode and Collector Emitter Junction of Transistor Keyed Open	K	Keyline	K	Keyed 5 ma Max Sink Current Keyed 20.5 VDC thru 10,000 ohms	The Keying Circuits Shall Operate Properly with the following Limit Inputs: Keyed-GND thru 15,000 ohms Max Keyed-50,000 ohms Min to GND
	U	+26.5 VDC (nom)	L	Direct Connections to +26.5 Primary Supply Line	+22 to 30 VDC
Direct Connection to Chassis GND	A	GND	M	Direct Connection to Chassis GND Inside PA	
Off-Open, 200,000 ohms Min On-GND thru C-E Junction of Transistor	T	PA On-Off Switching	N	Off +26.5 VDC thru 10,000 ohms On-Current 8 MA Max	The On-Off Circuits Shall Operate properly with the following Limit Inputs: On-GND thru 400 ohms Max Off 50,000 ohms Min to GND
Tune Start-GND Pulse thru Collector-Emitter Junction GND Pulse Width 30 ms Min Series Resistance 200 ohms Max Tune Start 26 VDC Nom thru 12 K ohms	C	Tune Start	P	Tune Start-13 ma Max Sink Current Tune Start 5.0 VDC thru 10,000 ohms and Diode (Amplifier) or +26.5 VDC thru 5700 ohms and Diode (with antenna coupler)	The Tune Start Circuits Shall Operate Properly with the following Limit Inputs: Tune Start-2 Milliseconds Min GND Pulse Thru 300 ohms Max Tune Start 50,000 Min to GND
Load Impedance 50,000 ohms Min	J	ALC/Transmitter Control	R	Positive Going Analog of RF Power Output Source Impedance Less than 2,000 ohms adjustable Level.	
	M	Fault		Fault (Amplifier or Antenna Coupler) GND thru Collector-Emitter Junction of Negative Positive Negative Silicon Transistor and Blocking Diode Fault-Open Diode Isolated From 26 V.	
Direct Connection to Chassis GND	B	Shield GND	G	Direct Connection to Chassis GND Inside PA	
Ant CPLR "On" 26.5 VDC Ant CPLR "Off" Open	D	Ant CPLR On-Off Switching	F	Direct Connection to Antenna Coupler (See Table II)	+22 to +30 VDC Max for On-Only Switching

NOTE: $\bar{A}$ - Negation of A

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Table II - Amplifier to Antenna-Coupler Interface

CU-2064/GRC Antenna Coupler Line Condition/Status	MS3114H14-15P Pin Connections	Line Function	MH20F(M)D00 Pin Connections	AM-6545/GRC Amplifier Condition/Status	Operating Limits
CPlr on 10mA Max current sink CPlr on (Normal Rx-Tx)-open, diode isolated	A	Antenna Coupler ON/Off	A	Direct connection to pin "T" of the amplifier-receiver exciter connector (See Table I)	
Fault-GND through collector emitter junction of saturated silicon NPN transistor $\overline{\text{Fault}}$ - Open collector emitter junction	B	Fault	B	Fault- 5mA Max sink current $\overline{\text{Fault}}$-20.5V through 10K and diode	The fault sensing circuits shall operate properly with the following limit inputs: Fault - GND through 1000 ohms Max $\overline{\text{Fault}}$ - 5000 ohms min to GND
Ready -GND through collector-emitter junction of saturated NPN silicon transistor $\overline{\text{Ready}}$-Open collector emitter junction	C	Ready	C	Ready-2mA Max sink current $\overline{\text{Ready}}$-10 Vdc (NOM) through 1000 ohms	The ready sensing circuits shall operate properly with the following limit inputs: Ready - GND through 1000 ohms Max $\overline{\text{Ready}}$-5000 ohms min to GND
Tx inhibit-GND through collector emitter of saturated silicon NPN transistor Rx inhibit-open collector Tx inhibit cont'd emitter junction (normal tune)	D	Transmit Inhibit	D	Interlock-10mA max sink current Interlock +26.5 Vdc (NOM) Max through 2700 ohms	The key interlocking circuit shall operate with the following limit inputs: Interlock-GND through 50 ohms Max $\overline{\text{Interlock}}$-1000 ohms Min to GND
Direct connection to chassis GND	P	GND	E	Direct connection to chassis GND inside PA	
Tune start - 10mA Max current sink $\overline{\text{Tune start}}$ -+26.5Vdc through 5700 ohms and diode	F	Tune Start	F	Direction connection to Pin "F" of the amplifier-receiver/ exciter connector (See Table I)	
Positive going DC analog of RF power, forward	H	Forward Power	H	Loaded by 1000Ohms and the 500 micro-ampere meter movement	
Tx-10 mA sink current $\overline{\text{Tx}}$-10 Vdc through 1800 ohms and diode	K	Keyline	K	Tx-Gnd through collector emitter junction of saturated NPN silicon transistor $\overline{\text{Tx}}$ open	
DC-AC inverter load (One or two blowers operating) 4.0 Max current drain	L	Blower +26.5 Vdc	L	Switched on/off by the amplifier primary power relay. Off-open line to +26.5 Vdc(NOM) primary line, system off or Rx-only mode. On +26.5 Vdc (NOM) in system on normal Rx-Tx mode	+22 to 30 Vdc
1.6A Max current drain in tune cycle	M	+26.5 Vdc primary power	M	Direct connection to +26.5 Vdc (NOM) primary supply line	+22 to 30 Vdc
Direct connection to chassis GND	E	Shield GND	P	Direct connection to chassis GND inside PA	
Fan test -50mA Max current sink $\overline{\text{Fan test}}$-GRD through relay coil resistance of 700 ohms Max	N	Blower test	N		

Note: $\overline{\text{A}}$ negation of A

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(3) Transmit keyline - ground for transmit operation

(4) Tune start - ground pulse whenever a frequency digit (except 100 Hz) is changed or when the dc power is switched ON for the amplifier.

(b) The amplifier shall provide the following control signals to the receiver-exciter:

(1) +26.5 Vdc (nominal) primary power supply

(2) System dc ground

(3) Fault line - ground for normal receive-transmit operation, open when an amplifier or antenna coupler fault is present.

(4) Tune-in progress - ground through relay contacts to switch and hold the receiver-exciter in the tune mode of operation.

(5) ALC - A negative going (for increasing attenuation) dc control signal at a source impedance not exceeding 2,000 ohms.

(6) Shield ground - ground connection to the shield of the interconnecting cable.

3.3.13.2 Amplifier to antenna coupler interface.

(a) The amplifier shall accept and operate with the following control input signals from the antenna coupler:

(1) Transmit interlock - open for normal operation, ground when the antenna coupler is not ready to accept rf power input.

(2) Fault - ground when an antenna coupler fault is present, open normally

(3) Direct current analog of forward power

(4) Direct current analog of reflected power

(5) Ready ground when antenna coupler has tuned correctly.

(6) Fan Test - +26.5 Vdc nominal to energize antenna coupler external fan when power amplifier (pa) lamp/fan test switch is activated.

(7) Antenna coupler on/off switch (thru connection to receiver-exciter - See Table I).

(8) Primary Power - +26.5 Vdc - Direct connection to system primary power line.

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(b) The amplifier shall provide the following control signals to the antenna coupler:

(1) +26.5 Vdc (nominal) primary power supply, switched on-off in the amplifier.

(2) System dc ground

(3) Keyline - ground for transmit mode of operation.

(4) Tune Start - momentary ground pulse whenever a frequency digit (except 100 Hz) is changed or when dc power is switched ON (direct connection to receiver-exciter TUNE START line).

(5) Shield ground - ground connection to the shield of the interconnecting cable.

(6) Primary Power - +26.5 Vdc

(7) Blower Test

3.3.14 System interface. The amplifier-receiver-exciter and amplifier-antenna coupler control interface lines shall be in accordance with Tables I and II, respectively. All interface lines (except grounds) shall be radio-frequency interference (rfi) filtered before exiting the amplifier shield enclosure.

3.3.15 Fasteners. Fasteners shall be provided as follows.

3.3.15.1 Cover fastening. The amplifier cover shall be fastened with captive stainless steel screws.

3.3.15.2 Mechanical fasteners. Except for the captive cover and and module mounting screws, Phillips head screws shall be used in the amplifier for securing mechanical and electrical parts. Use of screws smaller than 4-40 shall be minimized.

3.3.15.3 Module fasteners. All modules shall be secured to the case-chassis with 8-32 stainless steel captive screws. Locating pins shall be used where necessary to assure proper mating of connectors. Harness shall be secured to chassis with cable clamps.

3.3.15.4 Fasteners to vehicular mount. The amplifier shall be fastened to the Tray, Mounting, electrical equipment, MT-4497/GRC-193, Screw - Clamp fasteners.

3.3.16 Printed circuit board (pcb) keying. All plug-in pcb's shall be provided with keyed connectors to prevent incorrect insertion.

3.3.17 Printed circuit board test points. All pcb's shall be provided with color coded test jacks to facilitate troubleshooting.

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3.3.18 Radio frequency connections. The following coaxial rf cables shall be used in the amplifier:

(a) RG-188A/U at signal levels below 20 W.

(b) RG-142B/U double shielded cable for high level (400 W) connections.

(c) Type N connectors.

3.3.19 Module connections. All connections to functional modules shall be made either by plug-in connectors or screw type wire terminations. No unsoldering of wires shall be necessary to remove a module from the amplifier chassis.

3.3.20 Interface seal. The case-chassis and cover interface seal shall be waterproof. A combination waterand rfi seal shall be used.

3.3.21 Cooling. Heat transfer from power dissipating components and modules to the external heat exchanger shall be by conduction. The external blower shall be thermostatically controlled to operate only when the heat sink temperature exceeds +55°C. Visual inspection of the finned heat exchanger for obstructions to air flow shall be possible without disassembly of the amplifier.

3.3.22 Connectors. One-part printed circuit connectors shall not be used.

3.3.23 Disabling high voltage supply. An internal switch, accessible when the amplifier cover is removed, shall be provided for disabling the +1600-Vdc power supply. Switching the +1600-Vdc poower supply OFF shall be instantaneous. Switching the +1600-Vdc power supply ON shall be instantaneous except during the 60-second time delay immediately after the amplifier is switched ON.

3.3.24 Bandswitching interlock. Transmit and turning shall be inhibited for the duration of the bandswitch motor operation supplying a fault (ground signal to the control circuits in the amplifier and receiver-exciter.

3.3.25 00 and 01 MHz digit interlock. Transmit operation shall be inhibited whenever the receiver-excited freqency selectors are in the 00 and 01 position. A fault signal shall be supplied to the control circuits of the amplifier and receiver-exciter. The bandswitch motor shall be deenerized, leaving the bandswitch wipers in the last position before switching to 00 or 01 frequency digits.

3.3.26 Primary dc power distribution and switching. The amplifier shall be the dc power distribution and switching center for the receiver-exciter and antenna coupler used as part of the GRC-193 vehicular high frequency (hf) radio set system. Switching shall be remote, by the controls on the receiver-exciter. The equipment energizing and power line connections shall be in accordance with Table III.

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TABLE III - DIRECT CURRENT POWER DISTRIBUTION AND SWITCHING

RECEIVER/EXCITER CONTROL SETTINGS		DIRECTOR CURRENT PRIMARY POWER CONNECTIONS				EQUIPMENT ENERGIZING CONDITION						
						ANTENNA COUPLER				AMPLIFIER		
DIRECT CURRENT POWER	RECEIVE ONLY	RECEIVER/ EXCITER	ANTENNA COUPLER BLOWER	ANTENNA COUPLER CIRCUITS	AMPLIFIER	RECEIVER/ EXCITER	CIRCUITS	BLOWER, INTERNAL	BLOWER, EXTERNAL	CIRCUITS	BLOWER, EXTERNAL	DC-DC CONVERTER
0 1/	0	1 2/	0	1	1	0	0	0	0	0	0	0
0	1	1	0	1	1	0	0	0	0	0	0	0
1	0	1	1	1	1	1	1	1	3/	1	3/	4/
1	1	1	0	1	1	1	1	0	3/	0	3/	0

1/ 0 = "OFF", DEENERGIZED, DISCONNECTED
2/ 1 = "ON", ENERGIZED, CONNECTED
3/ SHALL BE THERMOSTATICALLY CONTROLLED (SEE 3.5.2)
4/ SHALL BE DELAYED FOR 65 SECONDS NOMINAL AFTER DC POWER HAS BEEN SWITCHED "ON". (SEE 3.7.6)

NOTE: THE ENERGIZING OF AMPLIFIER CIRCUITS AND ANTENNA COUPLERS SHALL BE IMPOSSIBLE WHEN THE AMPLIFIER CIRCUIT BREAKER IS OFF (TRIPPED).

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3.3.27 Clearing of system faults. Clearing system faults shall be possible in the following manner:

(a) 00 and 01 frequency digit fault - by moving the 10- and 1-MHz frequency selector switches away from the 00 and 01 positions.

(b) The high voltage supply overload device shall reset automatically.

(c) All other amplifier or antenna coupler fault conditions - by moving any frequency selected switch other than 100-Hz switch.

After clearing of system fault protective interlocks, return-to-normal operation shall be possible unless the fault condition is a permanent one. Attempts to resume normal operation after a fault has occurred shall not cause further damage to the equipment if the original fault fault was of the permanent type.

3.3.28 Amplifier and system faults. The amplifier shall sense and cause a system fault condition (inhibit tune and transmit operation) whenever one of the following conditions exist:

(a) For the duration of the 60-second warm-up time

(b) The high voltage supply overload protection circuits have disabled the dc-dc converter.

(c) The amplifier has failed to tune properly and the minimum inductance end limit switch of the variable coil in the tube tank circuit has been activated.

(d) The amplifier and the antenna coupler have not tuned properly in less than 20 seconds.

(e) The antenna coupler has faulted either due to voltage standing wave ratio (vswr) exceeding 2:1 for 400 milliseconds in normal operation or because it has not tuned properly in less than 10 seconds.

(f) The bandswitch motor is operating.

(g) The frequency selection controls are in the 00- or 01-MHz position.

When any of the above fault conditions exist, the amplifier shall provide the following:

(a) A fault ground signal (through a saturated transistor) to the receiver-exciter.

(b) Visual indications on the front face of the amplifier fault light (red) ON, ready light (green) OFF.

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Faults as described by (b) through (e) and (g) above shall cause a latching of the fault signal that shall be reset by a rechannnel pulse. The latch shall be ON if the fault were to be removed by an unexplained action. The keyline function and all related actions shall be inhibited during a fault. This requirement parallels this function in the receiver-exciter.

3.3.29 Controls, receptacles and indicators. Controls, receptacles and indicators shall be provided as follows:

3.3.29.1 External. The following controls, receptacles and indicators shall be located on the front face of the case-chassis assembly:

(a) Radio frequency input connector (N)

(b) Radio frequency output connector (N)

(c) Direct current primary power connector UW2020M00 in accordance with MIL-C-12520.

(d) Exciter control connector MW20F(M) D00 in accordance with MIL-C-55181.

(e) Antenna coupler control connector conforming to MS3114E14-15PW.

(f) 50-A circuit breaker (thermal type).

(g) Push button switch lamp/fun test.

(h) Ready light (green) - shall be on when amplifier and antenna coupler are ready for normal receive-transmit operation.

(i) Fault light (red) -

1. PA fault - shall be on indicating amplifier fault

2. Antenna coupler fault - shall be on indicating antenna coupler fault.

(j) Warning (amber)

1. Tune - shall be on indicating amplifier and antenna coupler in tune condition.

2. Tube fault - shall be on indicating burnout and a tube filement.

(k) Test meter, function measured, selected by internal switch.

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(l) A 3-A sealed thermal pushbutton circuit breaker on the +26 V direct line conforming to MIL-C-5809. Fuses shall not be used.

(m) Transmit light (green) shall be on when the power amplifier is in the transmit mode of operation.

3.3.29.2 Internal. The following controls and indicators shall be located inside the amplifier enclosure and shall be visible and accessible when the cover is removed:

(a) +1600 Volt (V) ON light (amber) - shall be lit when +1600 V present. This light shall be extinguished when the high voltage is abnormally low (less than +1200 Vdc).

(b) Overload light (red) - shall be lit when the dc-dc converter has been stopped by a high voltage overload.

(c) Test meter switch (11-position, rotary type) connected to the external test meter. The switch positions shall be as follows (starting from the counterclockwise (ccw) end stop):

1. +26.5 V (nom) dc primary line voltage
2. Filament of tube V1
3. Filament of tube V2
4. +19 Vdc
5. -14 Vdc
6. High Voltage (+1600 Vdc)
7. Drive level to tube stage
8. V1 and V2 plate current
9. ALC voltage
10. Forward power (normal position when the amplifier is closed and in operation.
11. Reflected power

NOTE: Positions 1, 8, 9, 10, and 11 above shall be analog indications, i.e., actual levels shall be measured. Positions 2 through 7 above shall be GO-NO-GO center scale indications on the meter.

3.3.29.3 Bandswitch code indicators. Light emitting diode type indicator lights shall be provided on the code converter pcb to show the band position.

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3.3.29.4 Lamp testing provision. The front panel spring-loaded single-pole double-throw (spdt) switch shall have one pole for simultaneously checking external indicator lights. The other pole shall energize the amplifier and coupler fans.

3.3.30 Interlocks and protective provisions. In order to prevent damage to the amplifier circuits and improper system operation, the following protective features shall be incorporated:

3.3.30.1 Overtemperature protection. Overtemperature protection shall deenergize the amplifier circuits when the heat sink temperature exceeds +96 C. Resetting of the protective device shall be automatic.

3.3.30.2 Overvoltage protections. The amplifier shall be deenergized when the primary supply voltage exceeds 33 Vdc. Resetting shall be automatic.

3.3.30.3 Reverse polarity protection. It shall be impossible to energize the amplifier when the primary power supply connections are accidentally reversed from the normal negative ground.

3.3.30.4 Overload protection - primary dc supply. A 50-A circuit breaker of the push-button type shall be connected in series with the positive primary power line. The circuit breaker shall be accessible from the outside of the amplifier case for checking and resetting. Fuses shall not be used. A separate 3-A breaker of the push-button type shall be connected in series with the antenna coupler +26 V direct line.

3.3.30.5 Overload protection - high voltage supply. The high voltage dc-dc converter shall be deenergized when the current drawn from the +1600-Vdc supply exceeds 800 milliampers.

3.3.30.6 Overload protection - low voltage supplies. The low voltage regulators shall be short-circuit proof.

3.3.30.7 Tube warm-up. The high voltage dc-dc converter shall remain deenergized and transmit keying shall be impossible for 60 ±10 seconds after the primary dc power is applied to the amplifier.

3.3.30.8 Key interlock. Transmit operation (TX keying) shall be inhibited during normal receive/transmit and tune operation when the antenna coupler is not ready to accept an rf power input.

3.3.30.9 Tune start interlocks. It shall be impossible to start the automatic tune cycle under the following conditions:

(a) The +1600-Vdc supply is not present

(b) There is no rf input signal from the exciter

(c) The amplifier variable inductor is not in the maximum inductance setting.

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(d) An equipment (amplifier or antenna coupler) fault exists.

3.3.30.10 Tune recycling interlocks. The automatic tune cycle shall be interrupted and all control ciruits returned to the tune start condition under the following conditions:

(a) An amplifier or antenna coupler fault occurs

(b) A frequency selector digit switch other than 100 Hz is moved, generating a single rechannel pulse.

(c) The amplifier and antenna coupler tune cycles are not completed in less than 20 seconds.

The homing function shall be accomplished without rf even if drive is presented by the receiver-exciter.

3.3.30.11 Electrical overload protection. Electrical overload protection shall be in accordance with Requirement B of MIL-STD-454.

3.3.31 Accessibility. All modules and components shall be easily accessible for troubleshooting, and replacement. All modules, vacuum tube and plug-in pcb's shall be removable from the amplifier chassis without disturbing any other modules or parts. No special tools shall be required. Only standard blade and Phillips type screwdrivers shall be necessary to loosen the mounting screws and remove wire connections. Accessible module test points shall be provided for use with standard test equipment, authorized at direct and general support, to locate and replace faulty modules. Special purpose cables and interface devices shall be provided for use with standard commercial and automatic test equipment available at depot for overhaul and repair of end item and modules.

3.3.32 Human engineering. Human engineering shall conform to Requirement 62 of MIL-STD-454.

3.3.33 Interchangeability. Provisions for interchangeability are required and shall be in accordance with Requirement 7 of MIL-STD-454.

3.3.34 Safety. Personnel safety hazards shall be kept to a minimum in accordance with Requirement 1 of MIL-STD-454.

3.4 Reliabililty. Reliability of operation shall be considered of prime importance in the design and manufacture of the amplifier. The contractor shall employ all methods possible in the process of design, development and manufacture which will assure quality and maximum reliability consistent with the state-of-the-art. In the functional application of parts, adequate factors of safety shall be provided by suitable derating from the part specification values in order to assure high equipment reliability under all applicable service conditions. The design shall include all possible features which will result in reliable and stable operation and hence reduce the requirements for maintenance by highly skilled personnel.

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3.4.1 Reliability program. The contractor shall conduct a reliability program which will provide for, but not be limited to, the following reliability tasks as described in MIL-STD-785:

(a) Reliability organization

(b) Management and control

(c) Subscontractor and supplier reliability programs

(d) Program reviews

(e) Design techniques

(f) Reliability analysis

(g) Parts reliability

(h) Reliability critical items

(i) Effects of storage, self-life, packaging, transportation, handling and maintenance

(j) Reliability demonstration

(k) Failure data collection, analysis and corrective action

(l) Status reports

3.4.2 Numerical reliability requirement. The reliability shall be a 1,000 hour mean time between failure (MTBF). This shall be achieved under the following operating conditions:

(a) Ambient operating temperature: 50°C

(b) Vehicle mounted, ground exposed environment

(c) Duty cycle: 9 to 1 receive to transmit ratio

3.4.3 Automatic tuning system life. The automatic tuning system shall perform a total of 6,000 complete tune cycles without failure or degradation requiring replacement of components or adjustment.

3.5 Maintainability. Design critieria for the maintainability of the amplifier and support equipment shall be in accordance with MIL-STD-470, MIL-STD-471 and MIL-HDBK-472. The mean-time-to-repair (MTTR) for the replacement of defective modules shall be no more than 15 minutes. The MTTR for the equipment modules, and subassemblies at the depot levels shall be no more than 1 hour.

3.6 Performance. The amplifier shall be capable of the following performance.

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3.6.1 Operation. The amplifier shall operate in wheeled or tracked vehicles, either stationary or in motion, powered by the normal primary dc power source of the vehicle.

3.6.2 Compatibility. The amplifier shall be operationally compatible with the AN/PRC-47, AN/PRC-74, AN/GRC-106, AN/GRC-108, AN/GRC-109, AN/GRC-122, AN/GRC-142, AN/PRC-104, AN/USC-2, AN/USC-3, AN/ARC-98 and AN/ARC-714 Radio Sets.

3.6.3 Modes of operations. The amplifier shall transmit voice communications using ssb and compatible am (usb plus carrier) and cw morse communications at frequencies from 2 to 29.999 MHz. In addition, transmission of secure voice (with a suitable applique device) communication shall be possible.

3.6.4 Standard conditions. The following standard conditions shall be used to establish normal performance characteristics and for making laboratory bench tests:

Temperature	Room ambient
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input voltage	26.5 ±0.3 V

3.6.5 Service conditions. The amplifier shall be capable of operating under the following service conditions:

(a)	Ambient temperature range	-40°C to +55°C
(b)	Ambient humidity range	0 to 100 percent relative
(c)	Vibration	Method 514, Procedure VIII, Schedule A, Curve V, 5-55 Hz 2.5 g or 0.15 inches, double amplitude whichever is the lesser in MIL-STD-810
(d)	Shock	Method 516, Procedures I, II and V of MIL-STD-810
(e)	Submersion	Method 512, Procedure I of MIL-STD-810
(f)	High temperature	Method 501, Procedure II of MIL-STD-810, except that at Step 4 the temperature shall be +55°C
(g)	Low temperature	Method 502, Procedure I of MIL-STD-810

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(h)	Humidity	Method 507, Procedure IV of MIL-STD-810
(i)	Altitude	Method 500, Procedure I of MIL-STD-810
(j)	Salt fog	Method 509, Procedure I of MIL-STD-810
(k)	Dust	Method 510, Procedure I of MIL-STD-810
(l)	Fungus	Method 508, Procedure I of MIL-STD-810

3.6.6 Storage conditions. The amplifier shall be capable of storage under the following conditions without degradation:

(a)	Ambient temperature range	-62°C to +65°C
(b)	Altitude	Sea Level to 70,000 feet

3.7 Electromagnetic-interference suppression. The amplifier shall meet the interference control limits of MIL-STD-461, using the test methods of MIL-STD-462. Terms and definition shall be in accordance with MIL-STD-463.

3.8 Dimensions. The maximum dimensions of the amplifier shall be 16.45 inches long, 11.4 inches wide and 8.05 inches high, excluding handles and cover holddown protrusions.

3.9 Weight. The weight of the amplifier shall not exceed 55 pounds.

3.10 Treatment and painting. Treatment and painting of the amplifier shall be in accordance with MIL-T-704, type A; the finish coat shall conform to MIL-E-52798 (forest green).

3.11 Marking and identifications. Marking shall conform to MIL-M-13231 and the following. Panel markings for switch positions shall be accurately aligned with the switch detents.

3.11.1 Nameplate. The amplifier shall contain nameplates in accordance with MIl-E-16400 and MIl-P-15024.

3.11.2 Test instructions. A decal on the inside of the cover shall be provided with the following instructions.

(a) Caution notes.

(b) Test switch positions and normal meter indications.

3.12 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirments specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3)

(b) Production (see 4.4)

(c) Sample (see 4.5)

4.3 First article inspections. The ten first articles shall be subjected to tests indicated for first article in Table IV. The Government representative present will select which amplifier is to be subjected to each test.

4.3.1 First article test reports. After completion of the first article tests, the contractor shall prepare a test report and submit it for approval in accordance with the terms of the contract.

4.4 Production inspection. All amplifiers shall be subjected to the production tests indicated in Table IV.

4.5 Sample inspections. During the production phase, sample tests shall be conducted as indicated in Table IV. The following quantities shall be selected by the Government inspector for sample tests:

Quantity of Amplifiers Offered for Acceptance	Quantity to be Tested
First 10	1
Next 90	1
Next 100	1
	One for each additional 100 or fraction thereof

4.5.1 Acceptance. No block of amplifiers shall be accepted for delivery by the Government representative until all sample tests have been successfully completed for that block.

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TABLE IV - TESTS

TEST	REQT. PARA.	TEST PARA.	FIRST ARTICLE	PRODUCTION	SAMPLE
Warm-up Time	3.3.3.3	4.7.4.1	x	x	x
Input Power	3.3.5	4.7.4.4	x		x
Input Voltage Variation	3.3.5.3	4.7.4.5	x		x
Power Output	3.3.3	4.7.4.6	x	x	x
Input Impedance	3.3.5.2	4.7.4.2	x		x
Spurious Output	3.3.7	4.7.4.7	x	x	x
Amplitude Modulation Distortion	3.3.7(c)	4.7.4.9	x	x	x
Automatic Tuning	3.3.9	4.7.4.3	x	x	x
Receive Only	3.3.10	4.7.4.8	x	x	x
Safety and Protective	3.3.30 3.3.11 3.3.23 3.3.24	4.7.4.10	x		x
Mechanical Inspection	3.3	4.3.2	x		x
Reliability	3.4.	4.7.2	x		
Automatic Tuning System Life	3.4.3	4.7.2.5	x		x
Maintainability	3.5	4.7.3	x		
System Operational Inspection	3.20	4.3.1	x		x
Humidity	3.6.5	4.7.1.2	x		
Vibration	3.6.5	4.7.1.4	x		x
Shock, Bench Handling	3.6.5	4.7.1.5.1	x		x
Shock, Drop	3.6.5	4.7.1.5.2	x		x
Shock, Basic Design	3.6.5	4.7.1.5.3	x		x
High Temperature	3.6.5	4.7.1.1.2	x		x
Low Temperature	3.6.5	4.7.1.1.1	x		x
Altitude	3.6.5	4.7.1.3	x		x
Salt Fog	3.6.5	4.7.1.8	x		
Dust	3.6.5	4.7.1.6	x		
Fungus	3.6.5	4.7.1.9	x		
Electromagnetic Interference	3.7	4.7.5	x		
Submersion	3.6.5	4.7.1.7	x		x

NOTE: SEQUENCE OF SERVICE CONDITION TESTS SHALL BE IN ACCORDANCE WITH TABLE I OF MIL-STD-810 FOR GROUP IIj EQUIPMENT.

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4.6 Test plans. Test plans, describing in detail first article, production and sample testing, shall be prepared by the contractor in accordance with requirements of MIL-STD-831 and submitted to the contracting officer for approval prior to initiation of testing. Approval of the test plans by the contracting officer or his representative is required prior to commencement of testing. The test plans shall include all test procedures deemed necessary and practicable by the contractor to demonstrate compliance with this specifications.

4.7 Tests.

4.7.1 Service condition tests. When a test requires measurements to be made before, during or after the test, these measurements shall be accomplished as specified in the test plans of 4.6. For those service condition tests where electrical measurements are not specified, the following measurements shall be made, as a minimum, prior to performing the service condition test and at the conclusion of the service condition test.

(a) Power output as specified in 4.7.4.6.

(b) Intermodulation distortion as specified in 4.7.4.7.

The equipment will be operated at a minimum of eight frequencies as detailed in the test procedure in the ssb mode for the above measurements. For those service condition tests where electrical measurements are not specified and the equipment is required to meet specified performance during testing, as a minimum, the measurements shown above shall be made during the test.

4.7.1.1 Temperature. During the temperature tests, the amplifier shall be operated in ssb mode, except at the temperature extremes where all modes shall be measured.

4.7.1.1.1 Low temperature. The amplifier shall be subjected to low temperature test in accordance with Method 502, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -65°C which shall be maintained until temperature stabilization of the test item is reached. The low operating temperature (Step 4) shall be -40°C. The chamber air circulating fan shall be baffled to protect the amplifier from direct air flow. Make the measurements listed in 4.7.1 prior to, or at Step 1 and at Step 5.

4.7.1.1.2 High temperature. The amplifier shall be subjected to high temperature test in accordance with Method 501, Procedure II of MIL-STD-810, except at Step 4 the temperature shall be +55°C. The high operating temperature for the amplifier shall be +55°C (including the effects of sun load). Make the measurements listed in 4.7.1 as a minimum, prior to, or at, Step 1 and at Steps 8 and 10.

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4.7.1.2 Humidity. The amplifier shall be subjected to humidity test in accordance with Method 507, Procedure IV of MIL-STD-810. The amplifier shall be operated in all modes where measurements are required. Make the measurements listed in 4.7.1 as a minimum at Step 4, Step 5, Step 6 and Step 8.

4.7.1.3 Altitude. The amplifier shall be subjected to altitude test in accordance with Method 500, Procedure I of MIl-STD-810. Make measurements listed in 4.7.1.

4.7.1.4 Vibration. The amplifier shall be subjected to vibration test in accordance with Method 514, Procedure VIII, Schedule A, Curve Y, 5 to 55 Hz, 2.5 g or 0.15 inches double amplitude whichever is the lesser in MIL-STD-810.

4.7.1.5 Shock.

4.7.1.5.1 Bench handling. The amplifier shall be subjected to bench handling test in accordance with Method 516, Procedure V, of MIl-STD-810. Make measurements listed in 4.7.1.

4.7.1.5.2 Drop. The amplifier, packaged for shipment, shall be subjected to drop test in accordance with Method 516, Procedure II, of MIL-STD-810. The amplifier shall be inspected for damage after completion of test. Make measurements listed in 4.7.1.

4.7.1.5.3 Basic design. The amplifier, prepared for normal field use, shall be subjected to the basic design shock test in accordance with Method 516, Procedure I, Figure 516-1, amplitude a and duration c of Mil-STD-810. Make measurements listed in 4.7.1.

4.7.1.6 Dust. The amplifier shall be subjected to dust test in accordance with Method 510, Procedure I, of MIL-STD-810. Make measurements as listed in 4.7.1.

4.7.1.7 Submersion. The amplifier shall be subjected to submersion test in accordance with Method 512, Procedure I, of MIl-STD-810. No electrical measurments are required.

4.7.1.8 Salt fog. The amplifier shall be subjected to salt fog test in accordance with Method 509, Procedure I, of MIL-STD-810. Operation of the amplifier shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the amplifier surfaces which can be attributed to flaky, pitted, blistered or otherwise loosened finish of metal surfaces.

4.7.1.9 Fungus. The amplifier shall be subjected to fungus test in accordance with Method 50B, Procedure I, of MIL-STD-810. Operation of the amplifier shall not be required. At the conclusion of the exposure period, examination shall reveal no evidence of viable fungus or corrosion on any surfaces.

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4.7.2 Reliablility.

4.7.2.1 Reliability acceptance test. The contractor shall conduct a reliability acceptance test in accordance with Test Plan III of MIL-STD-781. The specified MTBF shall be 1,000 hours. The minimum number of amplifiers to be tested shall be four. Mimimum test time per amplifier shall be in accordance with MIL-STD-781. The reliability testing shall be conducted in the sequence specified in MIL-STD-781.

4.7.2.2 Reliability test conditions. The reliability test described above shall be conducted under the following conditions. Test Level C of MIL-STD-781 shall be the test environment. A nine to one receive to transmit duty cycle shall be used. Proper instrumentation shall be provided to insure recognition of failure. A failure is defined as the inability of an amplifier to perform its required function. This includes catastrophic failure as well as any event which causes departure from performance specified herein or the contractor's approved test procedure.

4.7.2.3 Rework and retest provisions. Rework and retest provisions shall apply only to reliability acceptance test. Once a reject decision has been reached, no rerun can be made of that test on the same or a new sample from the same lot unless positive action consisting of design change, modification or rework actually relating to observed failures has been applied to the lot. Any subsequent rerun must start at zero hours operating time.

4.7.2.4 Burn-in (debugging) periods. Every amplifier shall be subjected to a 24-hour burn-in period. The last 12 hours of burn-in shall be failure free. The burn-in shall be conducted at test level C of MIL-STD-781 and the amplifier shall be exercised and monitored in the same manner as during reliability testing. The failures shall be recorded and analyzed but shall not be considered relevant for computation of the amplifier MTBF.

4.7.2.5 Automatic tuning system life. During the 6,000 tune-ups, various frequencies and dummy loads simulating different antennas shall be used in conjunction with the antenna coupler. A 50-ohm load for the amplififer in lieu of the coupler will not be acceptable. No more than 25 retunes requiring a second cycle shall be allowed. No retune in excess of two attempts shall be permitted and, as such, constitute failure of this life requirement. Full power output shall be recorded for each tune cycle.

4.7.3 Maintainability. The required MTTR shall be demonstrated in accordance with MIL-STD-471. The requirements of 3.5 shall be met.

4.7.4 Electrical. Unless otherwise specified herein, tests shall be performed under standard conditions specified in 3.6.4 and the following:

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(a) Test frequencies:

Band	Frequency (MHz)
A	2.1500
	2.9950
B	3.1500
	3.9950
C	4.1500
	5.9950
D	6.1500
	7.9950
E	8.1500
	11.9950
F	12.1500
	15.9950
G	16.1500
	23.9950
H	24.1500
	29.9950

(b) Modes of transmission

- o ssb two-tone signal
- o am
- o cw

4.7.4.1 Warm-up time. The capability of the amplifier to go through the automatic tune cycle shall be tested after the 60 ±10 second warm-up delay when the amplifier is initially turned on.

4.7.4.2 Input impedance. The amplifier shall be keyed into transmit mode of operation and the input impedance measured at 2, 5, 10, 15, 20 and 29 MHz. The resistive and reactive components of the input impedance shall be measured and the resultant vswr computed. The requirements of 3.3.5.2 shall be met.

4.7.4.3 Automatic tuning. Automatic tuning shall be tested at 2.1500, 5.9950, 12.1500, 23.9950 and 29.9950 MHz. This test shall be performed utilizing the antenna coupler. The requirements of 3.3.9 shall be met. During test specified in 4.7.2.4, a minimum of 200 tune-ups shall be completed, no more than five requiring in excess of a second cycle.

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4.7.4.4 Input power. The input power to the amplifier shall be measured in all modes of operation. The requirements of 3.3.5 shall be met.

4.7.4.5 Input voltage variation. The output voltage of all dc supplies shall be measured in the transmit and receive modes of operation while the dc input voltage is varied from 22 to 30 Vdc. The requirements of 3.3.5.3 shall be met.

4.7.4.6 Power output. The power output shall be measured into a 50-ohm load in all modes of transmission. The requirements of 3.3.3 shall be met.

4.7.4.7 Spurious output. Intermodulation distortion products, harmonics, hum, noise and other spurious outputs shall be measured in the the ssb mode at rated power output. The requirements of 3.3.7 shall be met.

4.7.4.8 Receive only. A test shall be performed to determine that the amplifier circuits are properly bypassed in the receive only mode (amplifier deenergized). The requirements of 3.3.10 shall be met.

4.7.4.10 Safety and protection. Tests shall be performed to determine the proper operation of all warning, protective and interlock functions. The requirements of 3.3.11, 3.3.23, 3.3.24, and 3.3.30 shall be met.

4.7.5 Electromagnetic interference test. The amplifier shall be subjected to test defined in MIL-STD-461 equipment type 1B.

4.8 Inspection.

4.8.1 System operational inspection. The amplifier shall be used in a complete vehicular system consisting of items in 3.3 and a suitable antenna to communicate with another identical system. The amplifier shall be checked for proper operation in all operating modes.

4.8.2 Mechanical inspection. The amplifier shall be given a thorough mechanical and visual inspection to determine that the quality of all materials and workmanship is in compliance with the requirements specified herein and MIL-E-16400. Particular attention shall be given to the following:

(a) Completeness

(b) Nameplates, identification markings and labels

(c) Ease of operation of connector mating, thumb screws, controls and switches and any other part intended for adjustment, movement or operation.

(d) Finishes

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(e) Welded joints

(f) Check of solder joints

(g) The fit of components in their rspective positions

(h) Check of mounting means

(i) Check of safety features

(j) Fastening and securing devices or parts

(k) Accessibility of components and parts for servicing

(l) Cable run between components including plugs and receptacles

(m) Fit of sealing gasket on cover and wear on gasket

(n) Check of all connector fittings

(o) Overall dimensions

(p) Weight

(q) Other visual defects

5. PACKAGING

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be the level of shipment specified in the contract or order in accordance with MIL-E-17555 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The amplifier is to be used as a part of the AN/GRC-193 fixed Radio Set, and the AN/MRC-138 vehicular system Radio Set to establish and maintain medium and long range hf communications. The equipment will be used in all types of weather and terrain including amphibious and airborn landing operations.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Acquisition

(a) Title, number and date of this specification

(b) Level of preservation, packaging and packing (see 5.1)

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements Lists (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirements	Applicable DID No.	Option
3.4.1	Reliability Program Plan	DI-R-7079	
4.3	First Article Inspection Procedures	DI-T-4901	
4.3.1	First Article Inspection Report	DI-T-4902	
4.7.2.1	Reliability Test Plan	Di-R-7033	
4.7.2.1	Reliability Test and Demonstration Report	DI-R-7034	

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.3 First article instructions. The first article shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. Selected first article tests will be witnessed by a representative of the procuring activity. Tests not witnessed by a representative of the procuring activity shall be witnessed and certified by the appropriate Defense Contract Administration Services District representative. Approval of the first article shall be by the contracting officer.

6.4 Marginal notations. Asterisks are not used this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - MC
Project No. 5820-N762

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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